

Conjecture 00000004011 is false for linear-growth vector fields:
the linear RDE $dY = Y dX$ with a Gaussian driver has lognormal,
not Gaussian, tails

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1 The conjecture and our reading

Statement. *Definition: An RDE is a rough differential equation $dY = V(Y) dX$. Conjecture: The tail of the solution norm satisfies the Fernique-type bound $P(\|Y\| > t) \leq 2 \exp(-t^2/c)$, where the constant c depends only on the growth constant of the vector fields and the tail parameters of the driver.*

Reading. We read “growth constant of the vector fields” as the constant K in the linear growth condition $|V(y)| \leq K(1 + |y|)$. Linear-growth RDEs are a standard class in the integrability literature: Friz and Riedel (arXiv:1104.0577) write in their abstract “Integrability properties of (classical, linear, linear growth) rough differential equations (RDEs) are considered”, and their paper has a “Remark 5 (Unbounded vector fields)”. The claim is: there is $c > 0$, determined by K and by the tail parameters of the driver, such that $P(\|Y\| > t) \leq 2 \exp(-t^2/c)$ for all $t \geq 0$. We refute it for one equation with $K = 1$ and one Gaussian driver, so no choice of the function $(K, \text{tail parameters}) \mapsto c$ can work. We also refute the weaker version that asks for the bound only for $t \geq T$, for any T .

Reading not refuted. If “growth constant” is read as a uniform bound $\sup_y |V(y)|$ (bounded vector fields), then $V(y) = y$ is out of scope and this note says nothing. In that setting Gaussian tails are expected: Friz and Riedel recall that “we recover the well-known fact that solutions of the Stratonovich SDE have Gaussian tails at any fixed point provided” the vector fields are sufficiently smooth.

The counterexample.

- Vector field $V(y) = y$ on $\mathbb{R}$. It satisfies $|V(y)| \leq 1 \cdot (1 + |y|)$ (growth constant 1) and is 1-Lipschitz.
- Driver $X_t = tG$ on $[0, 1]$, where G is a standard Gaussian random variable on a probability space (Ω, P) , i.e. $P \circ G^{-1} = N(0, 1)$. Each sample path is smooth, and $|X_t - X_s| = |G| |t - s|$, so the sup norm, the p -variation norm and the Hoelder seminorms of X on $[0, 1]$ all equal $|G|$. The driver has Gaussian tails: $P(|G| > t) \leq 2 \exp(-t^2/2)$ for $t \geq 0$. So its tail parameters are those of a standard Gaussian, and the driver itself satisfies the Fernique-type bound with $c = 2$.
- Initial value $Y_0 = 1$.

RDEs with smooth drivers. Rough path theory extends classical controlled differential equations: Wikipedia (“Rough path”) says “In stochastic analysis, a rough path is a generalization of the classical notion of a smooth path” and “This makes it possible to define and solve controlled

differential equations of the form” $dy_t = f(y_t)dx_t$. In its version of “Lyons’ Universal Limit theorem”, $X(n)$ is “a sequence of paths with finite total variation” and $Y(n)$ is “the solution to the differential equation” $dY(n) = V(Y(n))dX(n)$ in the classical sense; the rough-path solution is the limit of these classical solutions. For a driver that is C^1 on $[0, 1]$ we therefore use the classical equation

$$Y(0) = y_0, \quad Y'(t) = V(Y(t))X'(t) \quad (t \in [0, 1]),$$

with one-sided derivatives at 0 and 1 (Lean: `HasDerivWithinAt` on $[0, 1]$, with X' given by `derivWithin X (Icc 0 1)`). This is the definition `IsRDESolution`.

Conventions. For $0 < \alpha \leq 1$ the α -Hoelder *seminorm* of the driver equals $|G|$; the full Hoelder norm (sup norm plus seminorm) equals $2|G|$, which changes only a fixed constant in the driver’s Gaussian tail. For an arbitrary functional N whose tail events need not be measurable, the general Lean conclusion is an outer-measure statement; for the concrete sup norm, $N(Y) = \max(1, e^G)$ is measurable and the statement is an ordinary probability bound.

The solution norm. The text does not fix $\|Y\|$. We allow any path functional N with $Y_1 - Y_0 \leq N(Y)$ on the solution paths. This includes the sup norm on $[0, 1]$ (proved in Lean), the terminal value $|Y_1|$ (because $Y_0 = 1 > 0$), and the p -variation and Hoelder norms on $[0, 1]$, each of which is at least $|Y_1 - Y_0|$.

Conventions. $N(0, 1)$ is Mathlib’s `gaussianReal 0 1`, with density $\varphi(x) = (2\pi)^{-1/2}e^{-x^2/2}$. Probabilities are $[0, \infty]$ -valued measures and $2\exp(-t^2/c)$ enters as `ENNReal.ofReal`. The tail event is the strict one $\{N(Y) > t\}$. The probability space is arbitrary; we only assume that G is measurable with law $N(0, 1)$. Measurability of $\omega \mapsto N(Y_\omega)$ is not assumed (the lower bound uses only monotonicity of the measure).

2 Proof

Lemma 1 (existence and uniqueness). *For every $g \in \mathbb{R}$, $t \mapsto e^{tg}$ solves $Y' = YX'$, $Y_0 = 1$ with $X_t = tg$ on $[0, 1]$. Every solution Y satisfies $Y(t) = e^{tg}$ for $t \in [0, 1]$.*

Proof. Existence: $X' = g$ on $[0, 1]$ and $(e^{tg})' = ge^{tg}$. Uniqueness: $f(t) = Y(t)e^{-tg}$ has $f' = gYe^{-tg} - gYe^{-tg} = 0$ on $[0, 1]$, so f is constant, $f(t) = f(0) = 1$. $\square$

So for every ω , $Y_\omega(1) - Y_\omega(0) = e^{G(\omega)} - 1$, a shifted lognormal variable (e^G itself is lognormal).

Lemma 2 (Gaussian tail lower bound). *For $s \geq 0$, $P(G > s) \geq \varphi(s + 1)$.*

Proof. $P(G > s) \geq \int_s^{s+1} \varphi(x) dx \geq \varphi(s + 1)$, since φ is decreasing on $[0, \infty)$. $\square$

Theorem 3 (main). *Let Y_ω be any solution of $dY = YdX$, $Y_0 = 1$ with driver $X_t = tG(\omega)$, and let N satisfy $Y_\omega(1) - Y_\omega(0) \leq N(Y_\omega)$ for all ω . For every $c > 0$ and every $T \in \mathbb{R}$ there is $t \geq T$ with $P(N(Y) > t) > 2\exp(-t^2/c)$.*

Proof. Put $u = 1 + 20c + \max(T, 0)$ and $t = e^u - 1$. Then $u \geq 1$, $t \geq u \geq T$, and $t \geq u^2/2$ because $e^u \geq 1 + u + u^2/2$. If $G(\omega) > u$ then $N(Y_\omega) \geq e^{G(\omega)} - 1 > e^u - 1 = t$. Hence $P(N(Y) > t) \geq P(G > u) \geq \varphi(u + 1)$. It remains to show $2e^{-t^2/c} < (2\pi)^{-1/2}e^{-(u+1)^2/2}$. We have $u^2 > 20c$, $t^2 \geq u^4/4$ and $(u + 1)^2/2 + 3 \leq 5u^2$, so $((u + 1)^2/2 + 3)c \leq 5u^2c < u^4/4 \leq t^2$, i.e. $3 - t^2/c < -(u + 1)^2/2$. Since $\sqrt{2\pi} < 3$ and $e^3 \geq 1 + 3 + 9/2 > 6$, $2\sqrt{2\pi}e^{-t^2/c} < 6e^{-t^2/c} < e^{3-t^2/c} < e^{-(u+1)^2/2}$. $\square$

Remark 1. Taking $T = 0$: no $c > 0$ gives $P(N(Y) > t) \leq 2\exp(-t^2/c)$ for all $t \geq 0$. This is consistent with the literature: for linear RDEs, Friz and Riedel (arXiv:1104.0577, Remark 7)

note, in the case “which covers Brownian driving signals, we have log-normal tails. This is in agreement with trivial examples such as the standard Black-Scholes model in which the stock price is log-normally distributed.” The same mechanism works for the Brownian driver ($dY = Y \circ dB$, $Y_t = e^{B_t}$); we formalize only the smooth Gaussian driver.

3 Lean formalization

File `lean/Conjecture4011/Basic.lean` (namespace `C4011`, only import `Mathlib`, 281 lines). Definitions: `HasLinearGrowth` $\forall K (\forall y, |Vy| \leq K(1 + |y|))$, `IsRDESolution` $\forall X y_0 Y, \text{linDriver } g (t \mapsto tg)$, `supNorm01` $y (\sup_{t \in [0,1]} |y(t)| \text{ as an sSup})$. Theorems:

- `linearGrowth_id` : `HasLinearGrowth id 1`; `lipschitz_id_field`; `linDriver_increment`: $|X_t - X_s| = |g||t - s|$.
- `exp_isRDESolution`, `isRDESolution_unique` (Lemma 1).
- `gaussian_tail_lower` (Lemma 2); `driver_tail`: $P\{t < |G|\} \leq 2e^{-t^2/2}$ for $t \geq 0$ (Chernoff bound via `Mathlib`’s `HasSubgaussianMGF.measure_ge_le`).
- `tail_not_bernique` (Theorem 3): for Ω , P , measurable G with `P.map G = gaussianReal 0 1`, any Y with $\forall \omega, \text{IsRDESolution id (linDriver (G } \omega)) 1 (Y \omega)$ and any N with $\forall \omega, Y \omega 1 - Y \omega 0 \leq N (Y \omega)$: $\forall c, 0 < c \rightarrow \forall T, \exists t \geq T, \text{ENNReal.ofReal } (2 * \exp (-t^2/c)) < P \{\omega \mid t < N (Y \omega)\}$.
- `no_bernique_constant`: under the same hypotheses, $\neg \exists c, 0 < c \wedge \forall t, 0 \leq t \rightarrow P \{\omega \mid t < N (Y \omega)\} \leq \text{ENNReal.ofReal } (2 * \exp (-t^2/c))$.
- `no_bernique_constant_supNorm`: the same with $N = \text{supNorm01}$.
- `concrete_instance`: on $(\mathbb{R}, N(0, 1))$ with $G = \text{id}$ and $Y_\omega(t) = e^{t\omega}$ all hypotheses hold and the sup-norm bound fails.

Axiom audit. `lake build` succeeds. `#print` axioms for `tail_not_bernique`, `no_bernique_constant`, `no_bernique_constant_supNorm`, `concrete_instance`, `driver_tail` and `isRDESolution_unique` gives `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.

Sources

- Wikipedia, “Rough path”, https://en.wikipedia.org/wiki/Rough_path (introduction and section “Universal limit theorem”).
- P. Friz, S. Riedel, “Integrability of (non-)linear rough differential equations and integrals”, arXiv:1104.0577, <https://arxiv.org/abs/1104.0577> (abstract; Remarks 5, 6, 7).